Notes: Fisman and Wei (JPE, 2004)

Tax Rates and Tax Evasion: Evidence from "Missing Imports" in China

Contents

1 Big picture

- **Question.** How responsive is tax evasion to tax rates? The paper studies whether higher import taxes lead importers to hide taxable trade from Chinese customs.
- **Setting.** The authors examine China's imports from Hong Kong at the six-digit Harmonized System product level, focusing mainly on 1998 because the tariff schedule was stable throughout that year.
- **Measurement idea.** If there is no evasion or measurement error, Hong Kong's reported direct exports to China should equal China's reported imports from Hong Kong for the same product. The difference between the two reports is interpreted as an *evasion gap*.
- **Main variable.** The paper defines

  $$\text{gap\_value}_k = \log(\text{Hong Kong exports}_k) - \log(\text{China imports}_k),$$

  so a larger value means more missing imports on the Chinese side.
- **Core result.** A one-percentage-point increase in the tax rate is associated with roughly a 3 percent increase in the evasion gap.
- **Mechanisms.** Evasion occurs through at least two channels: underreporting values and misclassifying higher-tax products as lower-tax similar products.
- **Policy message.** Tax rates affect not only legal import demand but also the compliance margin. At the average tax rate in the sample, the paper argues that China may already be on the wrong side of the revenue-maximizing side of the Laffer curve.

2 Incremental contribution of the paper

- **A direct measure of evasion.** Earlier empirical work often inferred evasion from currency demand or aggregate national-account discrepancies. This paper constructs a product-level evasion proxy using mirror trade statistics.
- **Clean variation in tax rates.** The study uses detailed product-level variation in Chinese tariff plus VAT rates, avoiding the income-tax problem that marginal tax rates vary mechanically with income.
- **A high-frequency product-level approach.** The unit of observation is a narrow product category, such as a specific type of machine tool, not an aggregate industry or economy-wide time series.
- **Mechanism decomposition.** The paper separates underreporting of values, underreporting of quantities, and misclassification across similar goods.
- **Use of related-product tax rates.** The authors introduce the average tax rate on similar products within the same four-digit HS category to detect relabeling incentives.
- **First-difference robustness.** The paper uses China's 1997 tariff reform to check whether changes in tax rates predict changes in missing imports.
- **Policy contribution.** The estimates imply that simply raising statutory tax rates may reduce reported imports and tax revenue when evasion is sufficiently elastic.

3 Data and measurement

3.1 Trade reporting data

- The trade data come from the World Bank's World Integrated Trade Solution database, which draws on the United Nations Comtrade database.
- The main sample compares Hong Kong-reported direct exports to China with China-reported imports from Hong Kong.
- The authors work at the six-digit HS product level because Hong Kong's exports and China's imports can be matched at that level.
- The original 1998 sample contains 5,113 six-digit products. After excluding missing trade reports and products without consistent tax rates, the main sample has 1,663 products.
- Hong Kong also reports indirect exports to China. Because China may misclassify some indirect trade as direct imports from Hong Kong, the authors drop products with very low direct export ratios in the benchmark sample.

Interpretation: The key empirical advantage is that the same trade flow is reported by two customs systems. The key measurement challenge is that Hong Kong is a re-export hub, so the paper must separate direct exports from indirect exports.

3.2 Tax variables

- The tax rate is the sum of the Chinese import tariff and the VAT applied to imports.
- Tariffs are originally available at the eight-digit HS level. The paper uses six-digit products for which the underlying eight-digit tariff rates are uniform.
- The VAT rate varies from 13 percent to 17 percent across products.
- The 1998 benchmark is useful because the tariff schedule did not change during that year.

Interpretation: The paper treats the statutory product tax rate as the incentive to evade. The larger the tax rate, the larger the gain from hiding value, hiding quantity, or relabeling the product.

3.3 Evasion gaps

- The value gap is the log difference between Hong Kong-reported export value and China-reported import value.
- The quantity gap is the log difference between Hong Kong-reported export quantity and China-reported import quantity.
- A large value gap but a smaller quantity gap suggests underreporting of unit values.
- A value or quantity gap that is related to the taxes on similar products suggests product misclassification.

Interpretation: The sign convention is important: the notes follow the main-text definition, where a larger gap means more missing imports in China's customs data.

3.4 Similar-product tax rates and exemptions

- Similar goods are defined as other six-digit products in the same four-digit HS category.
- The variable avg(tax_o) is the export-value-weighted average tax rate on those other similar products.
- If a product's own tax is high but similar products' taxes are low, importers have an incentive to misclassify the high-tax product into the lower-tax category.
- The exemption ratio measures the share of imports exempt from import tariffs at the six-digit level.

Interpretation: The similar-product tax rate lets the paper distinguish a simple hidden-value story from a relabeling story.

4 Models and empirical specifications

4.1 Benchmark evasion regression

The main regression is

$$\text{gap\_value}_k = \alpha + \beta\,\text{tax}_k + u_k.$$

- k indexes six-digit products.
- $\beta > 0$ means higher taxes are associated with a larger missing-import gap.
- The benchmark estimate is approximately $\beta = 2.93$.
- The standard errors are clustered by four-digit HS category.

Interpretation: The coefficient means that a one-percentage-point higher tax rate is associated with about a 3 percent larger evasion gap.

4.2 Misclassification specification

To detect relabeling across related goods, the paper estimates

$$\text{gap\_value}_k = \alpha + \beta_1\,\text{tax}_k + \beta_2\,\text{avg}(\text{tax\_o})_k + u_k.$$

- $\beta_1 > 0$ captures the incentive to evade when the product's own tax rises.
- $\beta_2 < 0$ captures the incentive to misclassify the product into lower-tax similar categories.
- The estimates show a positive own-tax effect and a negative similar-product-tax effect.

Interpretation: If a product's own tax is high, evasion rises. If taxes on similar products are also high, there is less opportunity to hide the product by misclassification.

4.3 Quantity versus unit-value regressions

The quantity specifications replace gap_value with

$$\text{gap\_qty}_k = \log(\text{Hong Kong export quantity}_k) - \log(\text{China import quantity}_k).$$

- Without the similar-product tax control, the quantity gap is not strongly related to the own tax rate.
- Once the similar-product tax control is added, the own-tax coefficient becomes positive and the similar-product coefficient becomes negative.
- This pattern supports misclassification rather than broad underreporting of physical quantities.

Interpretation: Importers seem to hide value and relabel products more than they simply make all physical quantities disappear.

4.4 Exemptions, first differences, and nonlinearity

- Exemption regressions interact tax rates with the exemption ratio. If imports are legally exempt, the incentive to evade should be lower.
- First-difference regressions use changes from 1997 to 1998:

$$\Delta\text{gap}_k = \alpha + \beta_1 \Delta\text{tax}_k + \beta_2 \Delta\text{avg}(\text{tax\_o})_k + u_k.$$

- Flexible functional-form regressions allow the tax slope to vary across tax quartiles.
- Robustness checks vary the cutoff for the direct export ratio.

Interpretation: These specifications address measurement concerns, omitted product characteristics, and the possibility that evasion responds especially strongly only after taxes become sufficiently high.

5 Identification and empirical design

5.1 What the design identifies

- The design identifies product-level associations between statutory tax rates and missing imports.
- The paper's strongest interpretation is that higher tax rates induce more evasion because the gap is measured using two independent customs reports of the same flow.
- The first-difference evidence supports the interpretation that changes in tax rates change evasion gaps.

Interpretation: This is not a randomized experiment, but the mirror-statistics design gives a more direct evasion measure than aggregate underground-economy proxies.

5.2 Threats and responses

- **Indirect exports through Hong Kong.** China may misclassify some re-exports as direct imports from Hong Kong. The paper addresses this by excluding products with very low direct export ratios and by checking alternative cutoffs.
- **Product heterogeneity.** High-tax goods may be intrinsically harder to measure. The first-difference analysis removes time-invariant product-specific factors.
- **Legal exemptions.** Evasion incentives differ when many imports are legally exempt. The paper directly controls for exemption ratios and their interactions with taxes.
- **Misclassification versus underreporting.** The similar-product-tax variable helps distinguish relabeling from hidden values or hidden quantities.

Interpretation: The empirical strategy is strongest when the same product is compared across two reporting systems and when changes in tax rates predict changes in reporting gaps.

6 Tables and figures

6.1 Figures 1 and 2: Direct exports and tariff variation

Read:

- Figure 1 shows the distribution of the direct export ratio. Many products have low direct export ratios, confirming that Hong Kong's re-export role is important.
- Figure 2a shows substantial variation in tariff rates across six-digit products.
- Figure 2b shows meaningful within-four-digit variation in tariff rates, which is important for identifying product misclassification among similar goods.

Economic message: The paper has enough variation in tax rates to study evasion, but the low direct export ratios make careful treatment of Hong Kong re-exports necessary.

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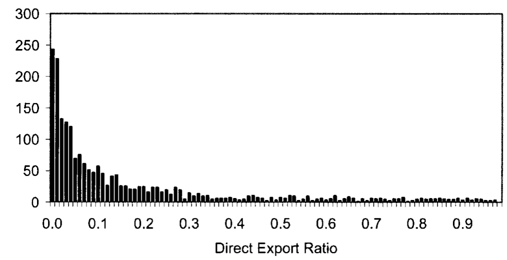

Fig. 1.—Frequency distribution of direct export ratio (see App. table A1 for a definition)

(a) Figure 1: Frequency distribution of direct export ratio

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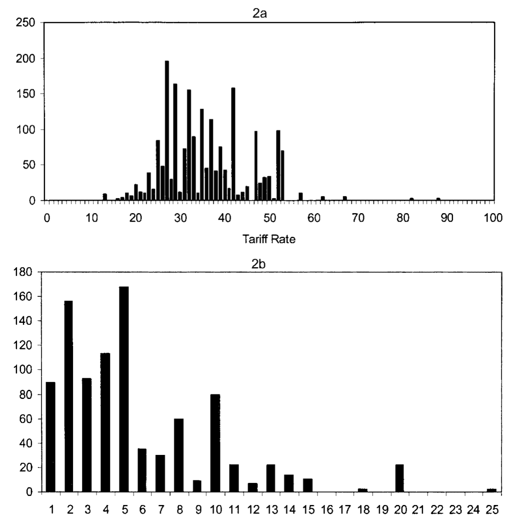

Fig. 2.—*a*, Frequency distribution of tariff rates by six-digit HS category. *b*, Frequency distribution of differences between maxima and minima of tariffs within each four-digit HS category.

(b) Figure 2: Tariff-rate variation

6.2 Tables 1 and 2: Tariff examples and sample description

Read:

- Table 1 gives a concrete example of tax-rate dispersion within a narrow product family: machine tools in HS category 8459.
- The tax rate differs sharply depending on the product subtype, especially whether the machinery is numerically controlled.
- Table 2 reports the main sample statistics. In the full sample, the average tax rate is about 36 percent and the average direct export ratio is 0.20.
- The evasion gap becomes positive when the sample is restricted to products with direct export ratios above the median, consistent with indirect-export misclassification depressing the raw gap.

Economic message: The data contain substantial tax variation even within related products, and the descriptive statistics show why the paper controls for Hong Kong's re-export role.

TABLE 1
TARIFF RATES AT THE SIX-DIGIT LEVEL FOR MACHINE TOOLS (Industry 8459)

HS Code	Industry Description	Tariff Rate (1998)
845910	Way-type unit head machines	35
845921	Other drilling machines: numerically controlled	26.7
845929	Other drilling machines: other	35
845931	Other boring-milling machines: numerically controlled	26.7
845939	Other boring-milling machines: other	37
845940	Other boring machines	31.9
845951	Milling machines, knee-type: numerically controlled	26.7
845959	Milling machines, knee-type: other	35
845961	Other milling machines: numerically controlled	26.7
845969	Other milling machines: other	37
845970	Other threading or tapping machines	35

(a) Table 1: Machine-tool tariff rates

TABLE 2
SUMMARY STATISTICS

	Mean	Standard Deviation	Minimum	Maximum	Observations
	A. Full Sample				
Log(value_export)	5.35	2.41	−4.51	12.72	1,663
Log(value_import)	5.47	2.50	−5.12	12.20	1,663
Gap_value	−.12	2.03	−7.95	9.64	1,663
Log(qty_export)	6.78	4.42	−2.30	20.17	1,199
Log(qty_import)	6.42	4.16	−2.30	21.99	1,603
Gap_qty	−.63	2.33	−10.16	11.74	1,102
Tax rate (tariff+VAT) (at the six-digit level)	.36	.09	.13	1.04	1,663
Avg(tax_o) (at the four-digit level)	.36	.09	.13	.88	1,470
Exemption ratio	.80	.29	.00	1.00	1,558
Direct export ratio	.20	.24	.01	.99	1,663
Change in tax rate, 1997–98	−.057	.060	−.30	.09	1,617
	B. Restricted to Products with Direct Export Ratio above the Median				
Log(value_export)	6.10	2.49	−4.51	12.72	839
Log(value_import)	5.53	2.68	−5.12	12.20	839
Gap_value	.57	2.01	−7.92	9.64	839
Log(qty_export)	8.18	4.47	−2.30	20.17	577
Log(qty_import)	7.01	4.34	−2.30	20.32	797
Gap_qty	.01	2.35	−8.63	11.74	526
Tax rate (tariff+VAT) (at the six-digit level)	.37	.10	.13	1.04	839
Avg(tax_o) (at the four-digit level)	.37	.9	.13	.67	746
Exemption ratio	.82	.29	.00	1.00	791
Change in tax rate, 1997–98	.063	.060	−.30	.082	828

(b) Table 2: Summary statistics

6.3 Tables 3 and 4: Raw evidence on higher taxes and larger gaps

Read:

- Table 3 compares median evasion gaps across high-tax and low-tax products and across products with minimum versus maximum taxes within four-digit categories.
- The gap is larger for products with higher tax rates, especially when within-four-digit tax dispersion is large.
- Table 4 cross-tabulates the same idea across and within four-digit categories.
- Both across-category and within-category comparisons point in the same direction: high-tax goods have larger missing-import gaps.

Economic message: Even before regression controls, the raw data are consistent with the central hypothesis: tax rates and missing imports move together.

TABLE 3
MEDIAN EVASION GAPS AND TAX RATES (across and within Four-Digit Classes)

A. DIFFERENT EVASION GAPS AT DIFFERENT TAX RATES (across Four-Digit Categories)

	Median Evasion Gap: Log(Export) − Log(Import) (1)
A. At low tax rates (below median, four-digit level)	−.16 (356 products)
B. At high tax rates (above median, four-digit level)	.23 (322 products)
C. difference in evasion = row B − row A	.39

B. DIFFERENT EVASION GAPS AT DIFFERENT TAX TATES (within Four-Digit Categories)

	Median Evasion Gap: All Four-Digit Categories (1)	Median Evasion Gap: Four-Digit Categories with Max(Tax) − Min(Tax) > .05 (2)
A. At the minimum tax rate (within a four-digit product line)	−.02 (293 products)	.14 (149 products)
B. At the maximum tax rate (within a four-digit product line)	.06 (385 products)	.54 (175 products)
C. Difference in gap: row B − row A	.08	.40

NOTE.—In both panels, entries reflect the median evasion gaps, or gap_value = log(reported exports) − log(reported imports), with the number of observations in parentheses. In panel A, row A represents observations on four-digit product lines for which the tax rates are below the median; row B represents observations on four-digit product lines for which the tax rates are above the median. Note that the numbers of observations between rows A and B in panel B differ because of multiple minima and maxima within each four-digit class.

(a) Table 3: Median evasion gaps and tax rates

TABLE 4
MEDIAN EVASION GAPS, CROSS-TABULATED BY DIFFERENT TAX RATES BOTH WITHIN AND ACROSS FOUR-DIGIT CATEGORIES

	Lower Tax Rates (1)	Higher Tax Rates (2)	Difference in Gap (Col. 2 − Col. 1) (3)
A. At the minimum tax rate (within four-digit categories)	−.23 (57 products)	.36 (92 products)	**.59**
B. At the maximum tax rate (within four-digit categories)	.25 (48 products)	.65 (127 products)	**.40**
C. Difference in gap: row B − row A	**.48**	**.29**	

NOTE.—The table is limited to four-digit categories in which maximum tax − minimum tax > .05. Entries reflect the median value of gap_value = log(reported exports) − log(reported imports), with the number of observations in parentheses. Col. 1 represents observations on four-digit product lines for which the tax rates are below the median; col. 2 represents observations on four-digit product lines for which the tax rates are above the median. Note that the

(b) Table 4: Cross-tabulated median gaps

6.4 Table 5 and Figure 3: Benchmark tax effect

Read:

- Table 5 estimates the benchmark value-gap regression.
- The coefficient on tax rate is about 2.9 in the full sample and remains close to 3 across outlier exclusions and subsamples.
- Figure 3 plots the average evasion gap by tax rate and shows a positive relationship, though with substantial noise.

Economic message: The benchmark estimate says that a one-percentage-point increase in tax rates is associated with roughly a 3 percent increase in missing imports.

TABLE 5
EFFECT OF TAX RATES ON EVASION (Measured in Value)

	REGRESSION						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Tax rate	2.93	2.46	3.21	3.57	2.98	2.61	3.4
	(.74)	(.67)	(.87)	(.89)	(.81)	(.79)	(.96)
Constant	−1.31	−1.04	−1.31	−1.48	−1.29	−1.12	−1.46
	(.29)	(.23)	(.30)	(.31)	(.29)	(.27)	(.34)
Excluding outliers?	no	yes	no	no	yes	yes	yes
Excluding products lacking tax on similar products?	no	no	yes	no	no	yes	yes
Excluding products lacking observations on quantities?	no	no	no	yes	yes	no	yes
Observations	1,663	1,639	1,470	1,102	1,087	1,450	968
R^2	.020	.017	.022	.031	.025	.017	.029

NOTE.—The dependent variable is log(value of exports from Hong Kong to China) − log(value of imports to China

(a) Table 5: Effect of tax rates on value evasion

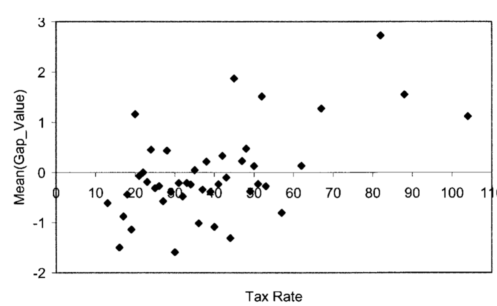

FIG. 3.—Relationship between mean(gap_value) and tax rates, 1998

(b) Figure 3: Mean gap and tax rates

6.5 Tables 6 and 7: Aggregation and misclassification

Read:

- Table 6 aggregates the evasion gap by tax brackets. The estimated tax coefficient remains close to the benchmark and the fit improves.

- Table 7 adds the average tax on similar products.
- The own-tax coefficient becomes larger, while the coefficient on the tax rate of similar products is negative.
- This pattern is consistent with misclassification: a high own tax raises evasion, but high taxes on similar alternatives reduce the ability to hide in lower-tax categories.

Economic message: The aggregate evidence supports the baseline result, and the similar-product-tax evidence shows that product relabeling is an important evasion channel.

TABLE 6
AGGREGATING THE EVASION GAP BY TAX BRACKETS

	DEPENDENT VARIABLE: Mean[Log(Value of Exports from Hong Kong to China) − Log(Value of Imports to China from Hong Kong)]*		DEPENDENT VARIABLE: Median[Log(Value of Exports from Hong Kong to China) − Log(Value of Imports to China from Hong Kong)]	
	(1)	(2)	(3)	(4)
Tax rate	2.90	2.41	3.29	2.56
	(1.17)	(.76)	(1.02)	(.89)
Constant	−1.17	−.97	−1.37	−1.101
	(.37)	(.28)	(.33)	(.32)
Method of aggregation	mean	mean	median	median
Weighting of data by number of observations per tax rate?	yes	no	yes	no
Observations	42	42	42	42
R^2	.23	.28	.30	.29

NOTE.—Robust standard errors are in parentheses.
* Means are taken for each level of tax rate.

(a) Table 6: Aggregated evasion gap by tax bracket

TABLE 7
INCORPORATING THE AVERAGE TAX ON SIMILAR PRODUCTS
Dependent Variable: Log(Value of Exports from Hong Kong to China) − Log(Value of Imports to China from Hong Kong)

	REGRESSION				
	(1)	(2)	(3)	(4)	(5)
Tax rate		6.07	5.31	8.32	7.46
		(1.37)	(1.25)	(1.56)	(1.42)
Tax on similar products	2.62	−3.16	−2.98	−4.65	−4.45
	(.90)	(1.39)	(1.33)	(1.58)	(1.53)
Constant	−1.09	−1.20	−1.02	−1.56	−1.33
	(.034)	(.31)	(.28)	(.38)	(.35)
Excluding outliers?	no	no	yes	no	yes
Excluding products lacking observations on quantities?	no	no	no	yes	yes
Observations	1,470	1,470	1,450	981	968
R^2	.014	.025	.020	.041	.035

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

Hong Kong–reported exports. We implement a regression of the following form:

(b) Table 7: Tax on similar products

6.6 Tables 8 and 9: Quantities and exemptions

Read:

- Table 8 uses the quantity gap rather than the value gap.
- The tax coefficient is weak without the similar-product tax control but becomes large and positive when that control is added.
- Table 9 controls for tariff exemptions.
- The exemption coefficient is negative, and the tax-exemption interaction implies that evasion incentives are much weaker when imports are legally exempt.

Economic message: Quantity evidence points toward misclassification rather than broad quantity disappearance, while exemption evidence confirms that evasion responds to actual tax liabilities.

TABLE 8
EVASION IN PHYSICAL QUANTITIES
Dependent Variable: Log(Quantity of Exports from Hong Kong to China) − Log(Quantity of Imports to China from Hong Kong)

	REGRESSION					
	(1)	(2)	(3)	(4)	(5)	(6)
Tax rate	1.13	1.14	.69		8.37	8.12
	(.93)	(1.11)	(.83)		(2.20)	(1.77)
Tax on similar products				.08	−7.93	−8.31
				(1.13)	(2.23)	(1.84)
Constant	−1.05	−1.08	−.92	−.68	−.85	−.65
	(.33)	(.40)	(.30)	(.40)	(.40)	(.36)
Excluding products lacking observations on avg(tax_*o*)?	no	yes	no	yes	yes	yes
Excluding outliers?	no	no	yes	no	no	yes
Observations	1,102	981	1,082	981	981	962
R^2	.003	.002	.001	.000	.015	.019

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

for products for which exemptions are rare, since exemptions provide

(a) Table 8: Evasion in physical quantities

TABLE 9
CONTROLLING FOR THE EFFECT OF TARIFF EXEMPTIONS

	DEPENDENT VARIABLE: Gap_Value				DEPENDENT VARIABLE: Gap_Qty			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Tax rate		2.96	5.30	16.69		1.05	7.07	18.73
		(.74)	(1.35)	(4.94)		(.89)	(2.22)	(6.19)
Tax on similar products			−2.20	−8.62			−6.61	12.16
			(1.40)	(4.83)			(2.32)	(6.03)
Exemption	−1.06	−1.02	−1.10	−1.42	−1.41	−1.37	−1.47	1.73
	(.23)	(.23)	(.25)	(.94)	(.31)	(.31)	(.33)	(1.16)
Exemption × tax rate				−15.34				−16.10
				(8.53)				(7.17)
Exemption × tax on similar products				8.91				8.54
				(5.85)				(7.02)
Constant	.72	−.40	−.41	−2.38	.48	.05	.30	−2.32
	(.23)	(.32)	(.38)	(.69)	(.31)	(.40)	(.46)	(.97)
Observations	1,558	1,558	1,362	1,362	1,028	1,028	905	905
R^2	.022	.043	.052	.066	.031	.033	.048	.061

NOTE.—Dependent variable: Cols. 1–4: log(value of exports from Hong Kong to China) − log(value of imports to China from Hong Kong); cols. 5–8: log(quantity of exports from Hong Kong to China) − log(quantity of imports to China from Hong Kong). Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

Consistent with the result that higher exemption rates lower the incentives for evasion, the coefficient on exemption is consistently negative

(b) Table 9: Controlling for tariff exemptions

6.7 Tables 10 and 11: First differences and nonlinear tax effects

Read:

- Table 10 uses changes from 1997 to 1998. Changes in tax rates predict changes in value gaps and quantity gaps.
- The first-difference estimates are noisier, but the signs match the level regressions.
- Table 11 allows the tax slope to differ across tax-rate quartiles.
- The effect is small at low tax rates and becomes much stronger once tax rates rise above the median.

Economic message: The paper's results are not only cross-sectional. Changes in tax rates also predict changes in evasion, and the compliance response is especially strong at higher tax rates.

TABLE 10
TAX AND EVASION IN FIRST DIFFERENCES, 1997–98

	DEPENDENT VARIABLE: Change in Gap_Value between 1997 and 1998		DEPENDENT VARIABLE: Change in Gap_Qty between 1997 and 1998	
	(1)	(2)	(3)	(4)
Change in tax rate	1.71	5.60	1.88	5.78
	(.85)	(1.92)	(1.48)	(2.91)
Change in tax on similar products		−3.97		−4.72
		(2.30)		(3.55)
Constant	.036	.01	.11	.05
	(.060)	(.063)	(.08)	(.091)
Observations	1,617	1,430	1,042	938
R^2	.004	.008	.002	.005

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

Flexible Functional Form

We now allow the marginal effect of a tax increase on evasion to differ

(a) Table 10: First differences, 1997–98

TABLE 11
EFFECT OF TAX RATES ON EVASION: FLEXIBLE FUNCTIONAL FORM
Dependent Variable: Log(Value of Exports from Hong Kong to China) − Log(Value of Imports to China from Hong Kong)

	REGRESSION	
	(1)	(2)
Tax rate in first quartile ($0 \leq$ tax rate < 29)	−2.72	−1.21
	(3.03)	(3.47)
Tax rate in second quartile ($29 \leq$ tax rate < 34)	.57	3.51
	(3.89)	(4.28)
Tax rate in third quartile ($34 \leq$ tax rate < 42)	6.33	9.54
	(2.88)	(3.43)
Tax rate in fourth quartile ($42 \leq$ tax rate)	3.12	6.51
	(1.52)	(2.49)
Average tax on similar products		−3.06
		(1.42)
Constant	.37	.77
	(.81)	(.83)
Observations	1,663	1,470
R^2	.025	.032

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

(b) Table 11: Flexible functional form

6.8 Tables 12 and 13: Robustness to direct-export-ratio cutoffs

Read:

- Table 12 repeats value-gap regressions under alternative cutoffs for the direct export ratio.
- Table 13 repeats the same robustness exercise for quantity gaps.
- The main patterns are qualitatively stable across cutoff values.
- The own-tax coefficient remains positive, and the average tax on similar products remains negative in the specifications that include it.

Economic message: The central results are not driven by the particular benchmark choice of excluding products with direct export ratios below 0.01.

TABLE 12
DIFFERING CUTOFFS FOR THE DIRECT EXPORT RATIO: EVASION IN VALUES
Dependent Variable: Log(Value of Exports from Hong Kong to China) − Log(Value of Imports to China from Hong Kong)

	Regression					
	(1)	(2)	(3)	(5)	(6)	(7)
Tax rate	2.82	3.10	4.03	6.32	5.27	5.11
	(.78)	(.75)	(.83)	(1.77)	(1.65)	(1.88)
Average tax on similar products				−3.57	−2.16	−1.31
				(1.70)	(1.71)	(2.06)
Constant	−1.63	−.86	−.96	−1.58	−.87	−.86
	(.27)	(.26)	(.30)	(.35)	(.30)	(.36)
Cutoff for (direct exports/total exports)	.00	.05	.10	.00	.05	.10
Observations	2,043	1,157	863	1,760	1,028	764
R^2	.015	.026	.041	.018	.029	.037

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

(a) Table 12: Direct-export-ratio robustness, values

TABLE 13
DIFFERING CUTOFFS FOR THE DIRECT EXPORT RATIO: QUANTITY REGRESSIONS
Dependent Variable: Log(Quantity of Exports from Hong Kong to China) − Log(Quantity of Imports to China from Hong Kong)

	Regression					
	(1)	(2)	(3)	(5)	(6)	(7)
Tax rate	.92	1.58	2.21	7.59	9.54	8.14
	(.84)	(.99)	(.83)	(2.08)	(2.72)	(2.79)
Average tax on similar products				−7.40	−9.14	−7.17
				(2.07)	(2.78)	(2.91)
Constant	−1.39	−.76	−.76	−1.17	−.38	−.35
	(.30)	(.37)	(.43)	(.41)	(.44)	(.50)
Cutoff for direct export ratio	.00	.05	.10	.00	.05	.10
Observations	1,368	770	592	1,191	692	534
R^2	.002	.006	.011	.011	.024	.017

NOTE.—Robust standard errors are in parentheses, accounting for clustering of standard errors by four-digit HS.

of imports and mislabeling higher-taxed products as lower-taxed

(b) Table 13: Direct-export-ratio robustness, quantities

6.9 Table A1: Variable definitions

Read:

- Table A1 lists the paper's main variables, including reported import values, reported export values, indirect exports, tax rates, exemptions, and the direct export ratio.
- The most important conceptual definitions are the value and quantity gaps, tax rate, exemption ratio, and average tax on similar products.
- For interpretation, use the main-text convention: a larger gap means more missing imports in China's customs records.

Economic message: The table clarifies that the empirical design is built from observable customs-reporting discrepancies, not from audit data or self-reported evasion.

TABLE A1
VARIABLE DEFINITIONS

Variable	Definition
Import_value	Value of imports (U.S. dollars) from Hong Kong into China, as reported by Chinese customs, for 1998, at the six-digit HS level; source: WITS derived from the United Nations' Comtrade database
Export_value	Value of exports (U.S. dollars) by Hong Kong destined for China, as reported by Hong Kong customs, for 1998, at the six-digit HS level; source: WITS
Indirect export_value	Value of indirect exports (U.S. dollars) from Hong Kong to China, originating from third-party countries; reported by Hong Kong customs, for 1998, at the six-digit HS level; source: WITS
Gap_value	log(import_value) − log(export_value)
Import_qty	Quantity of imports from Hong Kong into China, as reported by Chinese customs, for 1998, at the six-digit HS level (measured in U.S. dollars); source: WITS
Export_qty	Quantity of exports from Hong Kong destined for China, as reported by Hong Kong customs, for 1998, at the six-digit HS level (measured in U.S. dollars); source: WITS
Indirect export_quantity	Quantity of indirect exports, originating from third-party countries, as reported by the Hong Kong customs, for 1998, at the six-digit HS level (measured in U.S. dollars); source: WITS
Gap_qty	log(import_qty) − log(export_qty)
Tax	Total taxes rate levied on incoming goods by the Chinese authorities, equal to the sum of tariffs and commercial taxes, for 1998; source: WITS
*D*tax	Difference in the tax rates in 1998 and 1997 (see the text for details)
Exemption	At the six-digit level, the proportion of goods exempt from import tariffs; source: China Customs Statistics (purchased from the Economic Information Agency, Information Service Center, in Hong Kong)
Avg(tax_*o*)	Average of the level of tax for all other goods in a product's four-digit HS class, weighted by export_value
Direct export ratio	Ratio of direct exports to direct exports + indirect exports

Table A1: Variable definitions

7 Short summary

- The paper studies whether higher tax rates induce more tax evasion.
- The setting is China's imports from Hong Kong, using six-digit HS product-level trade reports.
- Evasion is proxied by the gap between Hong Kong-reported exports to China and China-reported imports from Hong Kong.
- The benchmark estimate implies that a one-percentage-point tax increase raises the evasion gap by about 3 percent.
- Misclassification is detected by adding the average tax rate on similar products: own taxes raise evasion while similar-product taxes reduce the relabeling opportunity.
- Quantity regressions suggest that the main mechanisms are value underreporting and product relabeling rather than simple disappearance of all physical quantities.
- Exemption and first-difference evidence support the interpretation that actual tax incentives drive the reporting gap.
- The effects are strongest at higher tax rates, implying that very high statutory rates may reduce reported imports and revenue.

8 Conclusion

8.1 Core conclusion

Fisman and Wei show that tax evasion responds strongly to tax rates. By comparing two customs reports of the same China-Hong Kong trade flows, they find that higher tax rates are associated with larger missing-import gaps.

8.2 Economic intuition

The intuition is straightforward. When the tax on a product rises, the importer gains more from hiding the product's value, underreporting taxable activity, or relabeling the product as a lower-tax similar good. The paper's strongest mechanism result is that misclassification matters: evasion is higher when a product's own tax is high and taxes on related products are low.

8.3 Incremental contribution revisited

The paper advances the tax-evasion literature by moving away from aggregate indirect proxies and using product-level mirror trade statistics. This allows the authors to estimate the tax responsiveness of evasion and distinguish mechanisms in a way that is difficult with income-tax audit data or macro-level underground-economy measures.

8.4 Policy implications

The findings imply that statutory tax rates and tax enforcement cannot be analyzed separately. If evasion rises strongly with tax rates, then higher rates may lower reported imports and reduce tax revenue. The paper's calculation suggests that the average Chinese import tax rate in the sample may already be sufficiently high that further increases could reduce revenue.

8.5 Limitations

The evasion gap may include measurement error, especially because Hong Kong is a major re-export hub. The paper addresses this with direct-export-ratio restrictions and robustness checks, but the reporting-gap measure is still an imperfect proxy for evasion. The analysis is also limited to imports from Hong Kong to China, so external validity to other trade routes and tax systems requires further study.

8.6 Takeaway

The main takeaway is that tax evasion is quantitatively important and tax-rate-sensitive. Higher tax rates do not simply raise revenue mechanically; they also increase the incentive to hide, understate, or relabel taxable trade.